

Cross-Field — Representative Vector

Domain-Partition Algorithm for Turbine Surfaces

Domain-Partition 3D Project

Block-Structured Meshing Pipeline

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Cross-Field $\rightarrow$ Representative Vector

- Each cross has 4 symmetric representatives (rotated by $\pi/2$)
- Pick the one closest to the desired direction
- Enables streamline integration along smooth field lines

Vector mapping:

$$\mathbf{v}(x) = \arg \max_{\mathbf{c} \in \text{cross}(x)} \langle \mathbf{c}, \mathbf{d}(x) \rangle$$

where $\mathbf{d}(x)$ is the desired direction.

Citation: Knöppel et al., "Globally optimal direction fields," SIGGRAPH 2013.